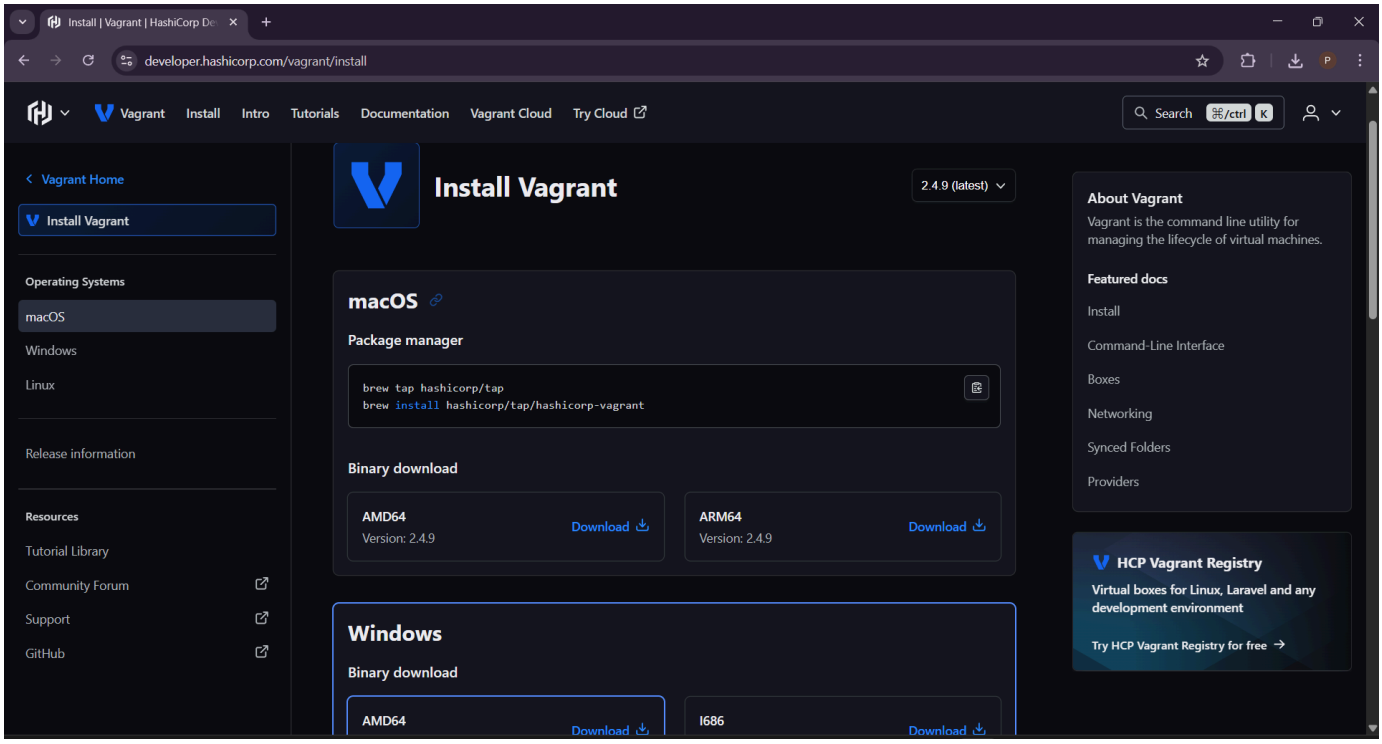
Cloud-containerization-and-DevOps

This is Experiment-1 Readme

Download Virtual Box from here

Download Vagrant from here



To verify the installation we will check the version via following command

```
vagrant --version
```

```
PS C:\Users\hp\Desktop\DEMO> vagrant -v
Vagrant 2.4.9
```

Initialize Vagrant with Ubuntu box:

```
vagrant init hashicorp/bionic64
```

```
Vagrant 2.4.9
PS C:\Users\hp\Desktop\DEMO> vagrant init hashicorp/bionic64
A `Vagrantfile` has been placed in this directory. You are now
ready to `vagrant up` your first virtual environment! Please read
the comments in the Vagrantfile as well as documentation on
`vagrantup.com` for more information on using Vagrant.
```

Start the VM:

```
vagrant up
```

```
PS C:\Users\hp\Desktop\DEMO> vagrant up
Bringing machine 'default' up with 'virtualbox' provider...
==> default: Box 'hashicorp/bionic64' could not be found. Attempting to find and install...
    default: Box Provider: virtualbox
    default: Box Version: >= 0
==> default: Loading metadata for box 'hashicorp/bionic64'
    default: URL: https://vagrantcloud.com/api/v2/vagrant/hashicorp/bionic64
==> default: Adding box 'hashicorp/bionic64' (v1.0.282) for provider: virtualbox
    default: Downloading: https://vagrantcloud.com/hashicorp/boxes/bionic64/versions/1.0.282/providers/virtualbox/unknow
n/vagrant.box
Progress: 65% (Rate: 21.5M/s, Estimated time remaining: 0:00:10)
```

Access the VM:

```
vagrant sshS
```

```
Windows PowerShell
PS C:\Users\hp\Desktop\DEMO> vagrant ssh
Welcome to Ubuntu 18.04.3 LTS (GNU/Linux 4.15.0-58-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/advantage

 System information disabled due to load higher than 1.0

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

0 packages can be updated.
0 updates are security updates.
```

Step 4: Install Nginx inside VM

```
sudo apt update
sudo apt install -y nginx
sudo systemctl start nginx
```

Verify Nginx

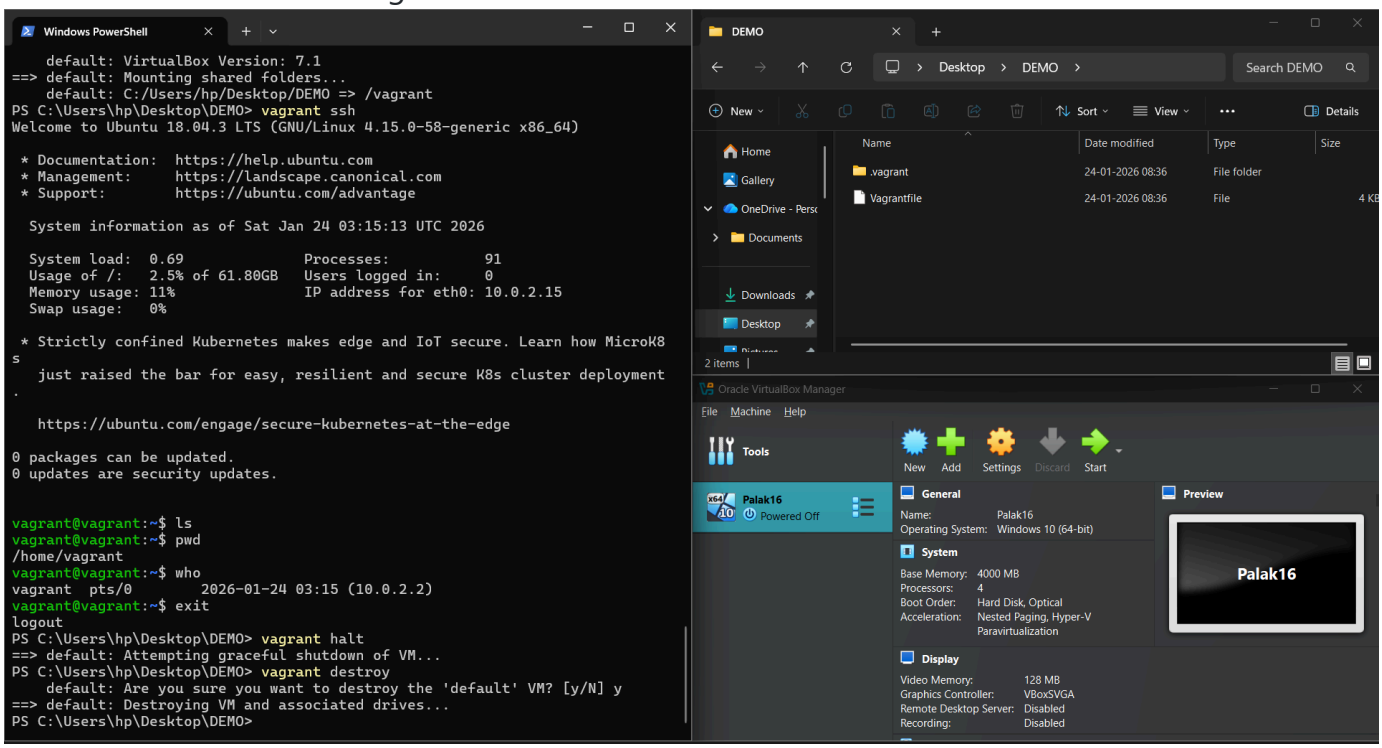
```
curl localhost
```

```
58261a837e072ada0319e55846b48ff294b3a02fba6e52bf7c3ec503b5cf2e98
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ curl localhost:8080
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```

Utilization Matrix In Running State

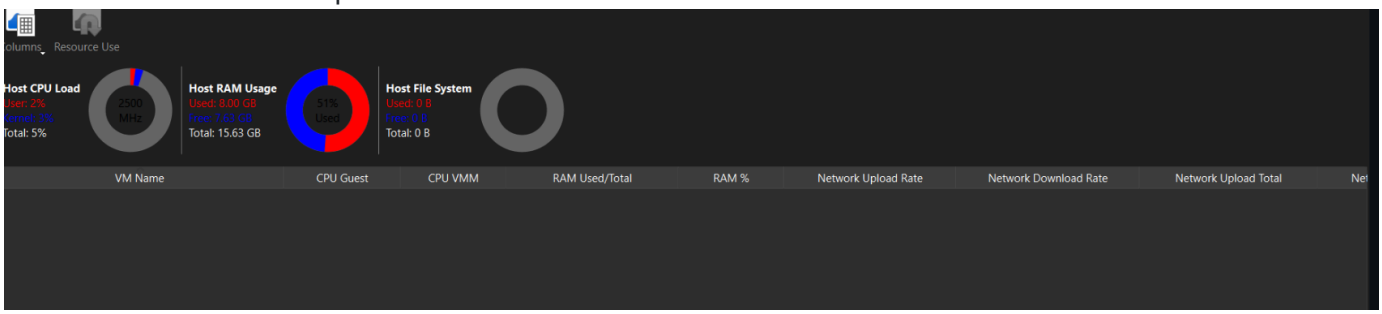


Stop VM

```
vagrant halt
```

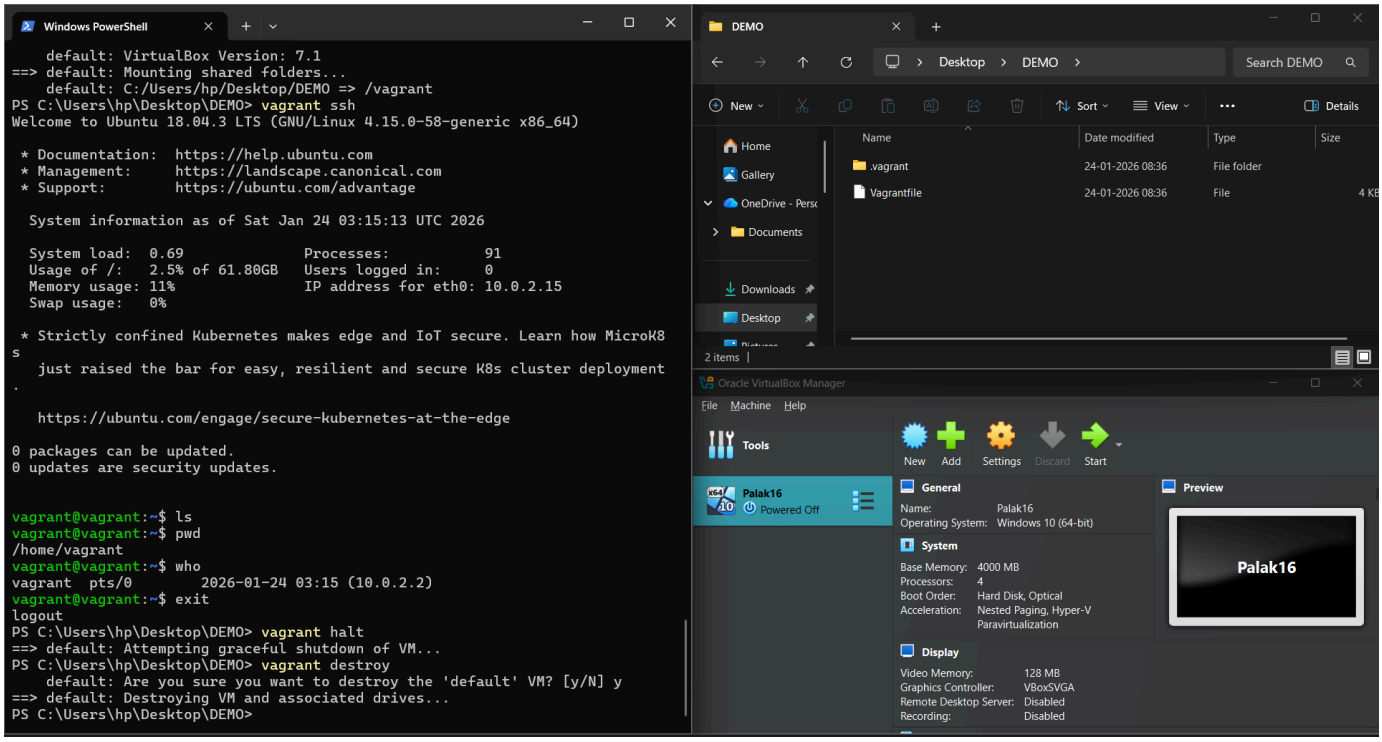
```
logout
PS C:\Users\hp\Desktop\DEMO> vagrant halt
==> default: Attempting graceful shutdown of VM...
PS C:\Users\hp\Desktop\DEMO> vagrant destroy
    default: Are you sure you want to destroy the 'default' VM? [y/N] y
==> default: Destroying VM and associated drives...
PS C:\Users\hp\Desktop\DEMO>
```

Utilization Matrix In Stop State



Remove VM

```
vagrant destroy
```



PART-B:- Containers using WSL (Windows)

Step 1: Install WSL

```
wsl --install
```

Reboot the system after installation.

Step 2: Install Ubuntu on WSL

```
wsl --install -d Ubuntu
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo --install -d Ubuntu
sudo: unrecognized option '--install'
usage: sudo -h | -K | -k | -V
usage: sudo -v [-ABkNnS] [-g group] [-h host] [-p prompt] [-u user]
usage: sudo -l [-ABkNnS] [-g group] [-h host] [-p prompt] [-U user]
            [-u user] [command [arg ...]]
usage: sudo [-ABbEHkNnPS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] [VAR=value] [-i | -s] [command [arg ...]]
usage: sudo -e [-ABkNnS] [-r role] [-t type] [-C num] [-D directory]
            [-g group] [-h host] [-p prompt] [-R directory] [-T timeout]
            [-u user] file ...
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
```

Step 3: Install Docker Engine inside WSL

```
sudo apt update
sudo apt install -y docker.io
sudo systemctl start docker
sudo usermod -aG docker $USER
```

Logout and login again to apply group changes.

Step 4: Verify Docker Installation

```
docker --version
```

```
exit
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ docker --version
Docker version 28.2.2, build 28.2.2-0ubuntu1~24.04.1
```

Step 5: Pull Ubuntu Image

```
docker pull ubuntu
```

```
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo apt install -y docker.io
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
docker.io is already the newest version (28.2.2-0ubuntu1~24.04.1).
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 86 not upgraded.
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo systemctl start docker
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo usermod -aG docker $USER
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:cd1dba651b3080c3686ecf4e3c4220f026b521fb76978881737d24f200828b2b
Status: Image is up to date for ubuntu:latest
docker.io/library/ubuntu:latest
```

Step 6: Run Ubuntu Container with Nginx

```
docker run -d -p 8080:80 --name nginx-container nginx
```

```
Run 'docker run --help' for more information
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ sudo docker run -d -p 8080:80 --name nginx-container nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
119d43eec815: Pull complete
700146c8ad64: Pull complete
d989100b8a84: Pull complete
500799c30424: Pull complete
10b68cfefee1: Pull complete
57f0dd1befe2: Pull complete
eaf8753feae0: Pull complete
Digest: sha256:c881927c4077710ac4b1da63b83aa163937fb47457950c267d92f7e4dedf4aec
Status: Downloaded newer image for nginx:latest
58261a837e072ada0319e55846b48ff294b3a02fba6e52bf7c3ec503b5cf2e98
```

Step 7: Verify Nginx in Container

```
curl localhost:8080
```

```
58261a837e072ada0319e55846b48ff294b3a02fba6e52bf7c3ec503b5cf2e98
palak@LAPTOP-2SDBVCPO:/mnt/c/Users/hp$ curl localhost:8080
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</head>
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<h1>Welcome to nginx!</h1>
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<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
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<p><em>Thank you for using nginx.</em></p>
</body>
</html>
```
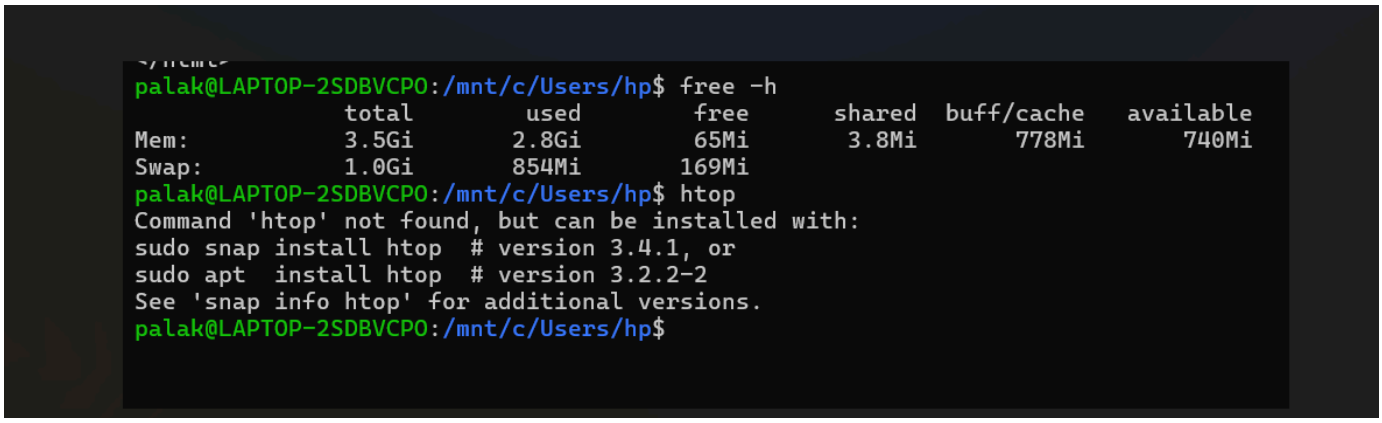

Step 8: Container Observation Commands

```
docker stats
free -h
```



Parameters to Compare

Parameter	Virtual Machine	Container
Boot Time	High	Very Low
RAM Usage	High	Low
CPU Overhead	Higher	Minimal
Disk Usage	Larger	Smaller
Isolation	Strong	Moderate